# Profiling_Better_Fusion.json

Server feedback (status/metrics/errors):


```
{
 ...
 "reference_runtime": 47.5,
 "kernel_runtime": 22.8,
 "speedup": 2.08,
 "metadata": {
  "hardware": "NVIDIA H100 80GB HBM3",
  "profiling": {
   "kernels": [
    {
     "name": "max_pool3d_k2s2_per_channel_kernel",
     "cuda_time_us": 96169.16500000001,
     "cpu_time_us": 0.0,
     "count": 5
    },
    {
     "name": "sm90_xmma_dgrad_implicit_gemm_...",
     "cuda_time_us": 6663.944000000018,
     "cpu_time_us": 0.0,
     "count": 5
    },
    {
     "name": "max_pool3d_k3s3_per_channel_kernel",
     ...
    },
    ...
    {
     "name": "sum_channels_kernel",
     ...
    ],
   "total_cuda_time_us": 118674.11,
   "total_cpu_time_us": 243259.05,
   "kernel_count": 8,
   "memory_stats": {
    ...
   }
  },
  ...
  "triton_profiler_matches": "[...'max_pool3d_k2s2_per_channel_kernel grid...',
                'max_pool3d_k3s3_per_channel_kernel...', 'sum_channels_kernel...']",
  "custom_kernel_cuda_time_in_profiling_us": 102242.61,
  "total_kernel_run_time_in_profiling_us": 118674.10,
  "custom_kernel_cuda_time_coverage": "Custom kernel CUDA time: 102242.61us / Total time:
118674.11us, Coverage: 86.15%",
  ...
```